

Part B - Applied Scenario (15 Marks)

Requirement: A single SQL query producing:

- department_name
- total_students in the department
- average_grade of the department
- number_of_courses offered by the department

Show only departments with at least 2 courses, ordered by average_grade in descending order.

Solution (single SQL query):

```
SELECT d.department_name,  
       COUNT(DISTINCT s.student_id) AS total_students,  
       ROUND(AVG(e.grade), 2) AS average_grade,  
       COUNT(DISTINCT c.course_id) AS number_of_courses  
FROM Departments d  
LEFT JOIN Students s ON s.department_id = d.department_id  
LEFT JOIN Enrollments e ON e.student_id = s.student_id  
LEFT JOIN Courses c ON c.department_id = d.department_id  
GROUP BY d.department_id, d.department_name  
HAVING COUNT(DISTINCT c.course_id) >= 2  
ORDER BY average_grade DESC;
```

Explanation:

- COUNT(DISTINCT s.student_id) counts unique students in the department.
- AVG(e.grade) calculates the average grade across all enrollments for students in that department.
- COUNT(DISTINCT c.course_id) counts the courses offered by the department.
- HAVING ensures only departments with at least 2 courses are shown.
- ORDER BY average_grade DESC sorts departments from highest to lowest average grade.